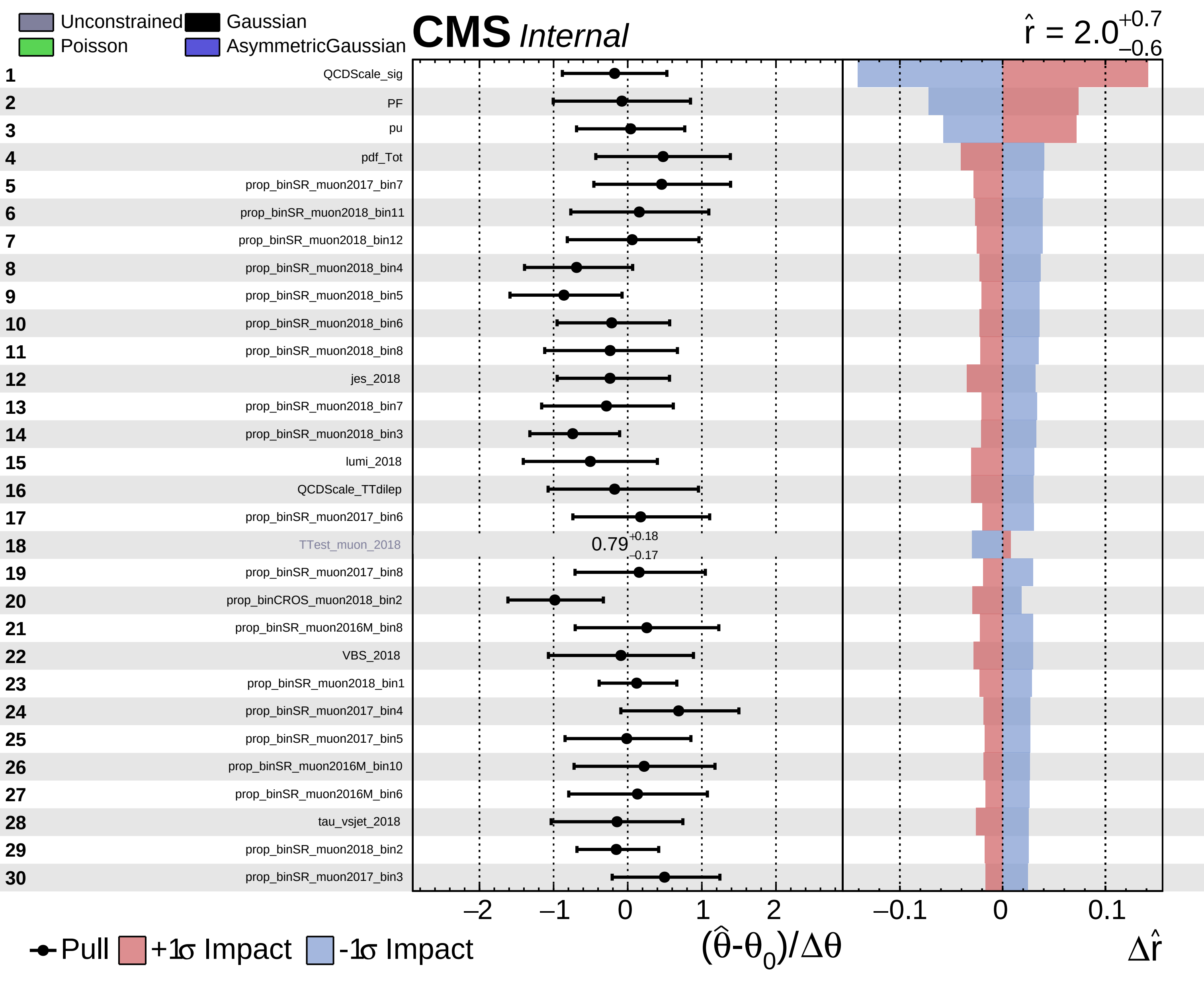
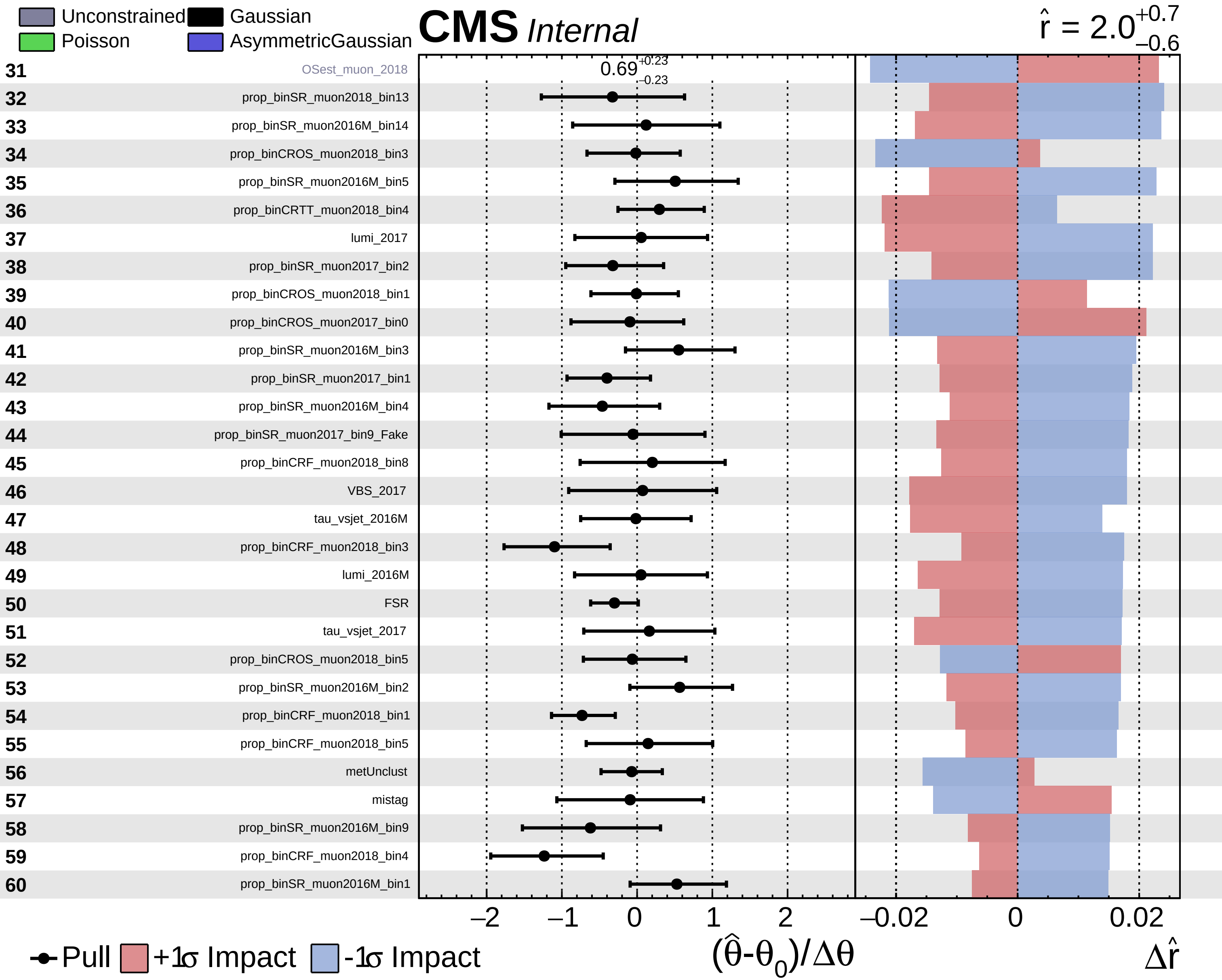
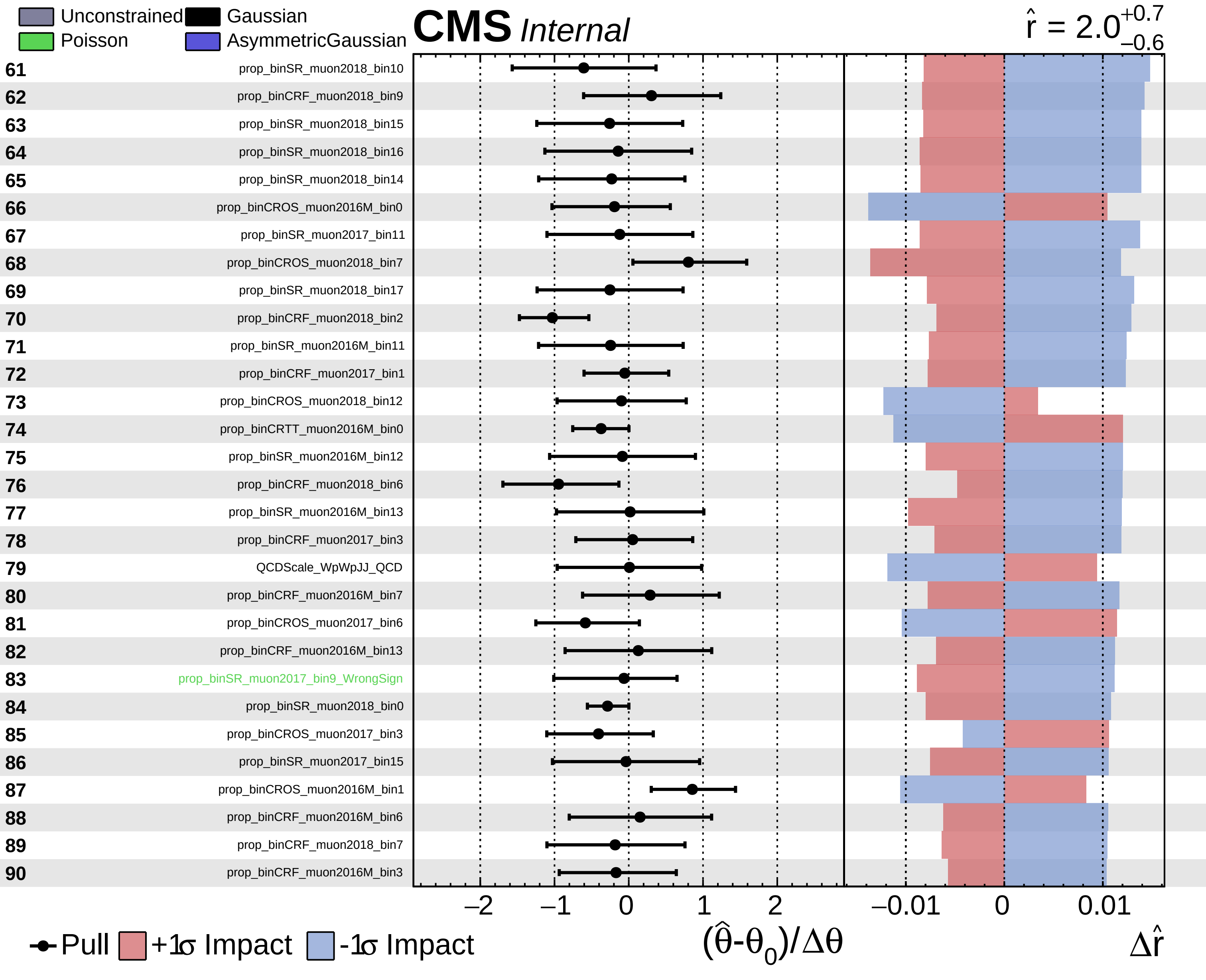
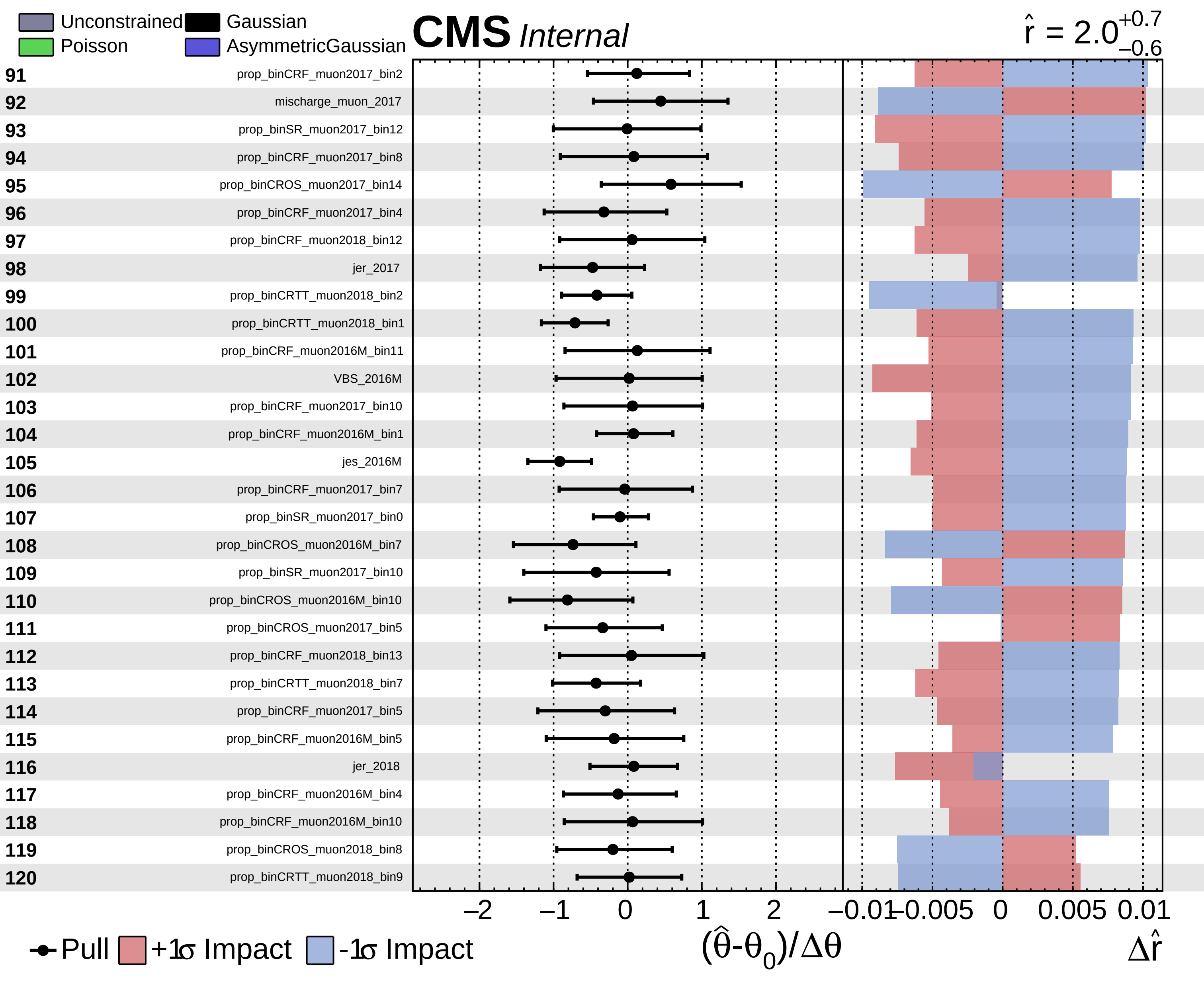


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
1 QCDScale_sig
2 PF
3 pu
4 pdf_Tot
5 prop_binSR_muon2017_bin7
6 prop_binSR_muon2018_bin11
7 prop_binSR_muon2018_bin12
8 prop_binSR_muon2018_bin4
9 prop_binSR_muon2018_bin5
10 prop_binSR_muon2018_bin6
11 prop_binSR_muon2018_bin8
12 jes_2018
13 prop_binSR_muon2018_bin7
14 prop_binSR_muon2018_bin3
15 lumi_2018
16 QCDScale_TTdilep
17 prop_binSR_muon2017_bin6
18 TTest_muon_2018
$0.79^{+0.18}_{-0.17}$
19 prop_binSR_muon2017_bin8
20 prop_binCROS_muon2018_bin2
21 prop_binSR_muon2016M_bin8
22 VBS_2018
23 prop_binSR_muon2018_bin1
24 prop_binSR_muon2017_bin4
25 prop_binSR_muon2017_bin5
26 prop_binSR_muon2016M_bin10
27 prop_binSR_muon2016M_bin6
28 tau_vsjet_2018
29 prop_binSR_muon2018_bin2
30 prop_binSR_muon2017_bin3
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

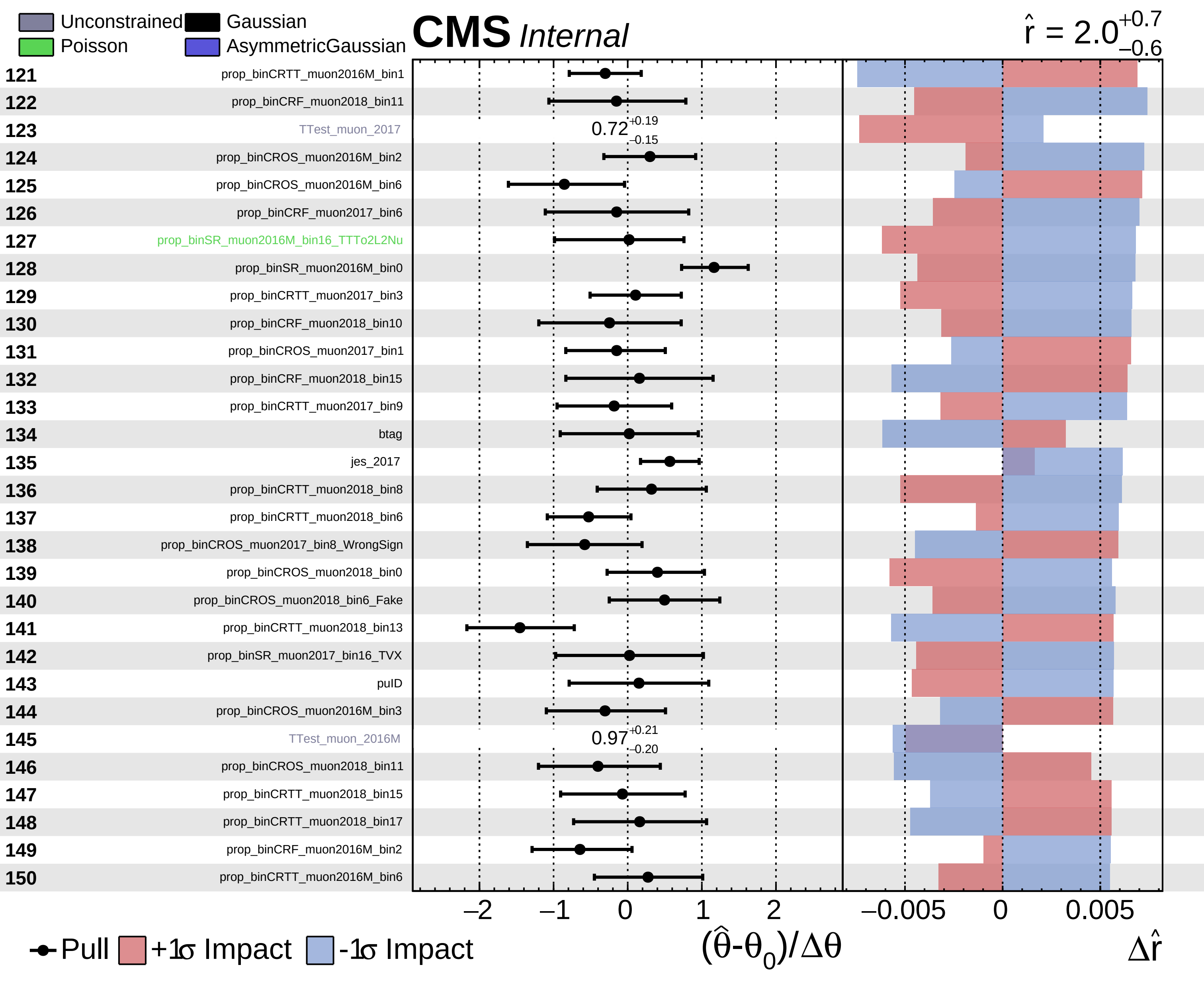

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
31 OSest_muon_2018
0.69 +0.23 −0.23
32 prop_binSR_muon2018_bin13
33 prop_binSR_muon2016M_bin14
34 prop_binCROS_muon2018_bin3
35 prop_binSR_muon2016M_bin5
36 prop_binCRTT_muon2018_bin4
37 lumi_2017
38 prop_binSR_muon2017_bin2
39 prop_binCROS_muon2018_bin1
40 prop_binCROS_muon2017_bin0
41 prop_binSR_muon2016M_bin3
42 prop_binSR_muon2017_bin1
43 prop_binSR_muon2016M_bin4
44 prop_binSR_muon2017_bin9_Fake
45 prop_binCRF_muon2018_bin8
46 VBS_2017
47 tau_vsjet_2016M
48 prop_binCRF_muon2018_bin3
49 lumi_2016M
50 FSR
51 tau_vsjet_2017
52 prop_binCROS_muon2018_bin5
53 prop_binSR_muon2016M_bin2
54 prop_binCRF_muon2018_bin1
55 prop_binCRF_muon2018_bin5
56 metUnclust
57 mistag
58 prop_binSR_muon2016M_bin9
59 prop_binCRF_muon2018_bin4
60 prop_binSR_muon2016M_bin1
−2
−1
0
1
2
−0.02
0
0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



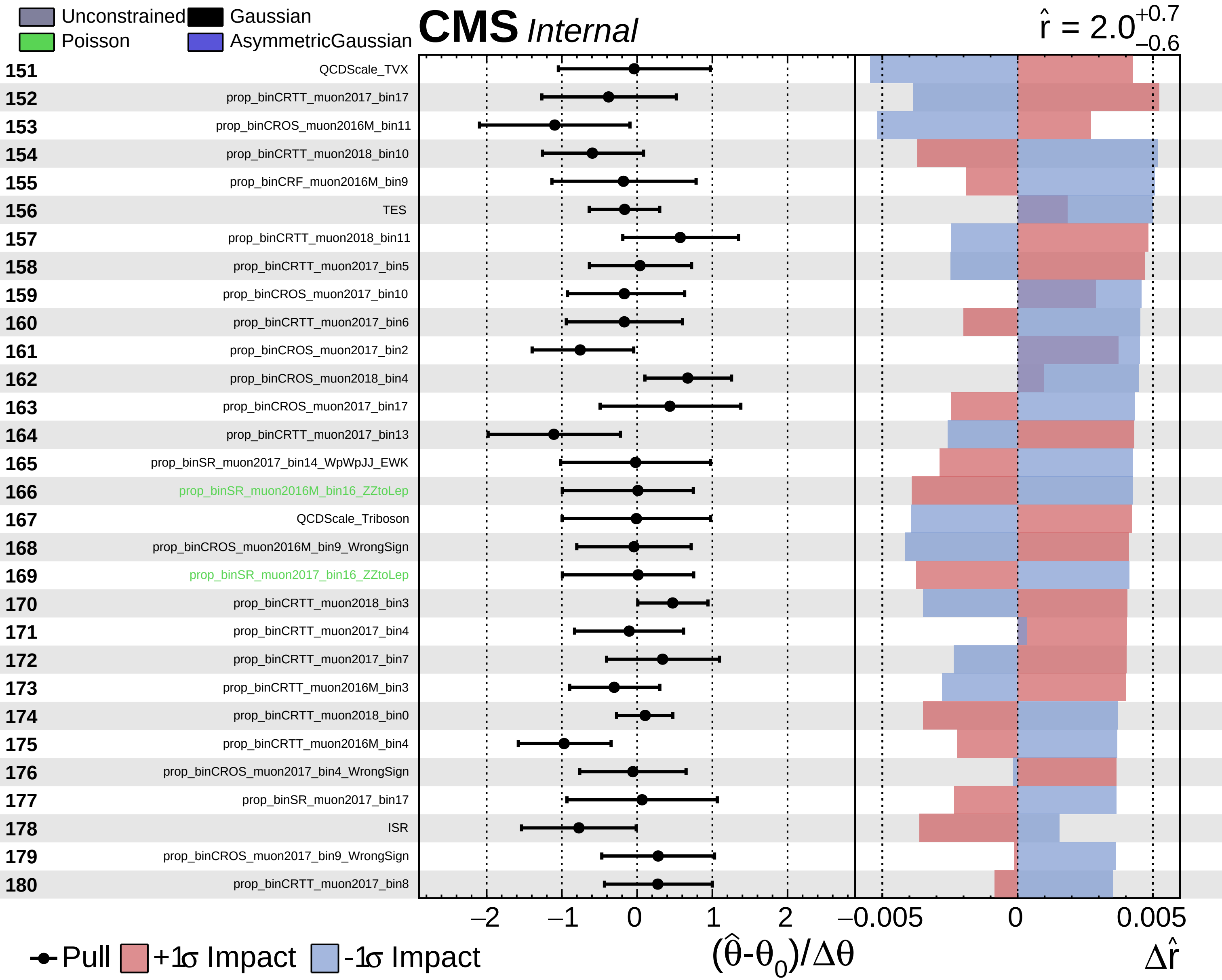
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.0 +0.7 −0.6
61 prop_binSR_muon2018_bin10
62 prop_binCRF_muon2018_bin9
63 prop_binSR_muon2018_bin15
64 prop_binSR_muon2018_bin16
65 prop_binSR_muon2018_bin14
66 prop_binCROS_muon2016M_bin0
67 prop_binSR_muon2017_bin11
68 prop_binCROS_muon2018_bin7
69 prop_binSR_muon2018_bin17
70 prop_binCRF_muon2018_bin2
71 prop_binSR_muon2016M_bin11
72 prop_binCRF_muon2017_bin1
73 prop_binCROS_muon2018_bin12
74 prop_binCRTT_muon2016M_bin0
75 prop_binSR_muon2016M_bin12
76 prop_binCRF_muon2018_bin6
77 prop_binSR_muon2016M_bin13
78 prop_binCRF_muon2017_bin3
79 QCDScale_WpWpJJ_QCD
80 prop_binCRF_muon2016M_bin7
81 prop_binCROS_muon2017_bin6
82 prop_binCRF_muon2016M_bin13
83 prop_binSR_muon2017_bin9_WrongSign
84 prop_binSR_muon2018_bin0
85 prop_binCROS_muon2017_bin3
86 prop_binSR_muon2017_bin15
87 prop_binCROS_muon2016M_bin1
88 prop_binCRF_muon2016M_bin6
89 prop_binCRF_muon2018_bin7
90 prop_binCRF_muon2016M_bin3
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

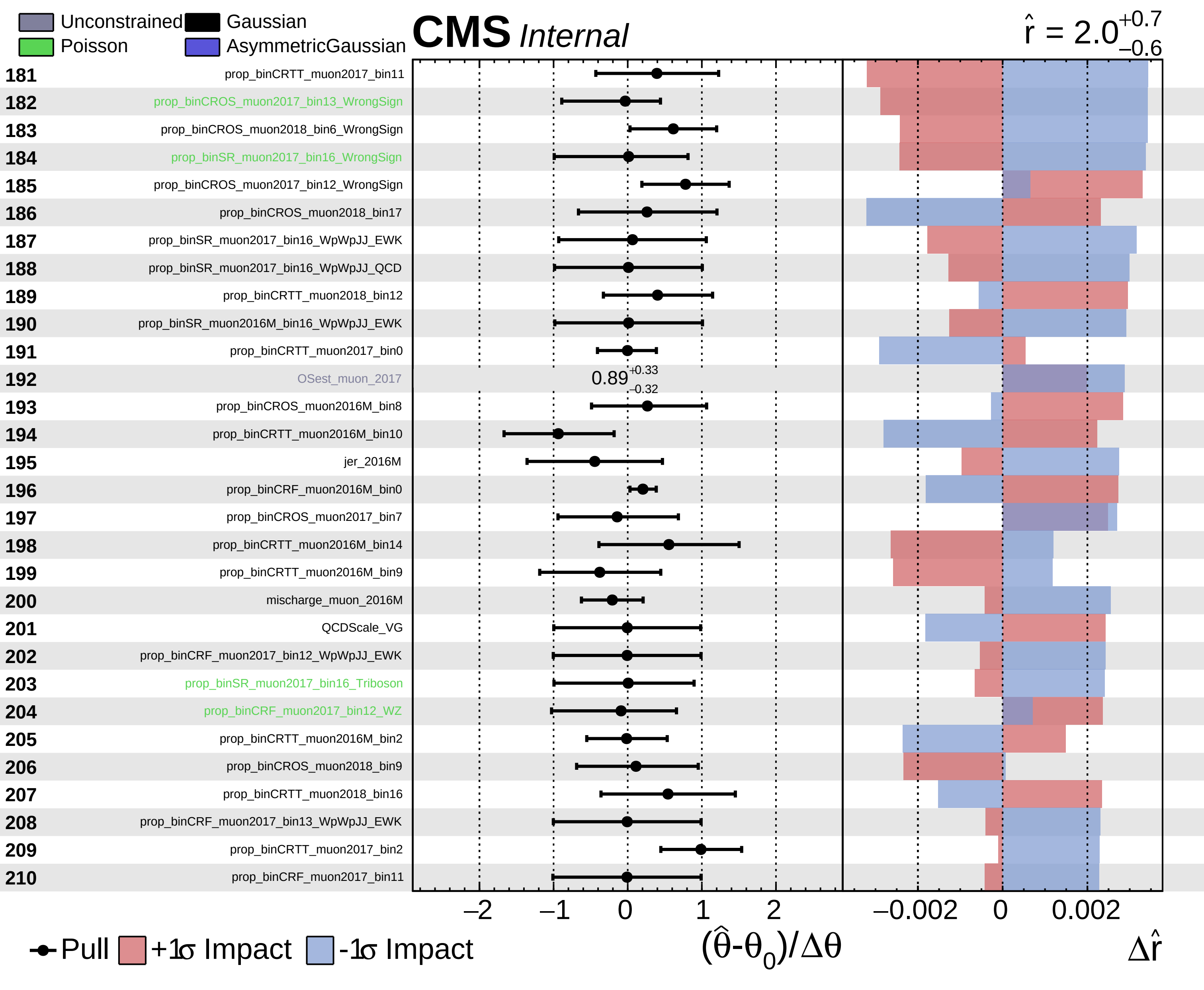


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
91 prop_binCRF_muon2017_bin2
92 mischarge_muon_2017
93 prop_binSR_muon2017_bin12
94 prop_binCRF_muon2017_bin8
95 prop_binCROS_muon2017_bin14
96 prop_binCRF_muon2017_bin4
97 prop_binCRF_muon2018_bin12
98 jer_2017
99 prop_binCRTT_muon2018_bin2
100 prop_binCRTT_muon2018_bin1
101 prop_binCRF_muon2016M_bin11
102 VBS_2016M
103 prop_binCRF_muon2017_bin10
104 prop_binCRF_muon2016M_bin1
105 jes_2016M
106 prop_binCRF_muon2017_bin7
107 prop_binSR_muon2017_bin0
108 prop_binCROS_muon2016M_bin7
109 prop_binSR_muon2017_bin10
110 prop_binCROS_muon2016M_bin10
111 prop_binCROS_muon2017_bin5
112 prop_binCRF_muon2018_bin13
113 prop_binCRTT_muon2018_bin7
114 prop_binCRF_muon2017_bin5
115 prop_binCRF_muon2016M_bin5
116 jer_2018
117 prop_binCRF_muon2016M_bin4
118 prop_binCRF_muon2016M_bin10
119 prop_binCROS_muon2018_bin8
120 prop_binCRTT_muon2018_bin9
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

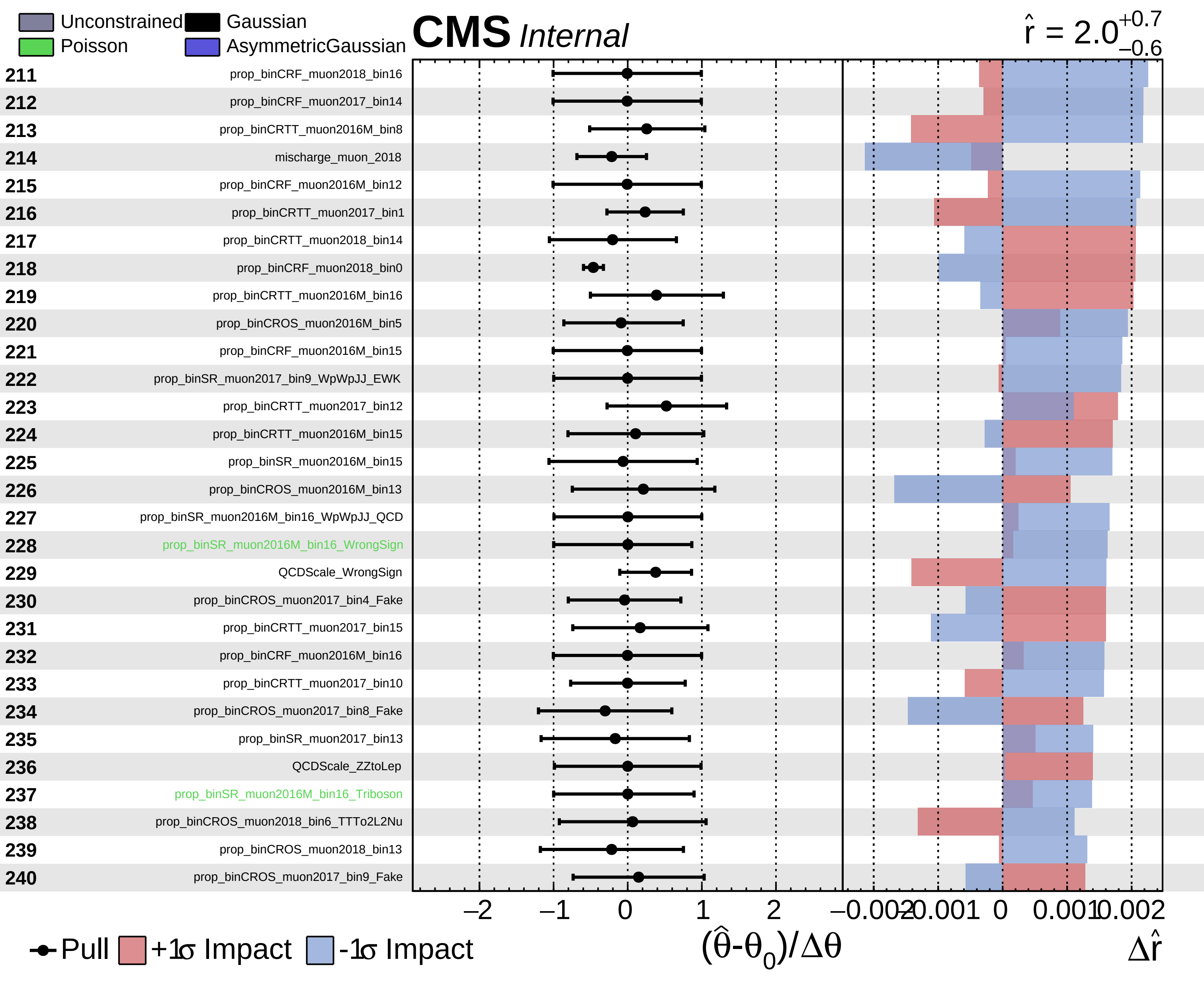


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
121 prop_binCRTT_muon2016M_bin1
122 prop_binCRF_muon2018_bin11
123 TTest_muon_2017
$0.72^{+0.19}_{-0.15}$
124 prop_binCROS_muon2016M_bin2
125 prop_binCROS_muon2016M_bin6
126 prop_binCRF_muon2017_bin6
127 prop_binSR_muon2016M_bin16_TTTo2L2Nu
128 prop_binSR_muon2016M_bin0
129 prop_binCRTT_muon2017_bin3
130 prop_binCRF_muon2018_bin10
131 prop_binCROS_muon2017_bin1
132 prop_binCRF_muon2018_bin15
133 prop_binCRTT_muon2017_bin9
134 btag
135 jes_2017
136 prop_binCRTT_muon2018_bin8
137 prop_binCRTT_muon2018_bin6
138 prop_binCROS_muon2017_bin8_WrongSign
139 prop_binCROS_muon2018_bin0
140 prop_binCROS_muon2018_bin6_Fake
141 prop_binCRTT_muon2018_bin13
142 prop_binSR_muon2017_bin16_TVX
143 puID
144 prop_binCROS_muon2016M_bin3
145 TTest_muon_2016M
$0.97^{+0.21}_{-0.20}$
146 prop_binCROS_muon2018_bin11
147 prop_binCRTT_muon2018_bin15
148 prop_binCRTT_muon2018_bin17
149 prop_binCRF_muon2016M_bin2
150 prop_binCRTT_muon2016M_bin6
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

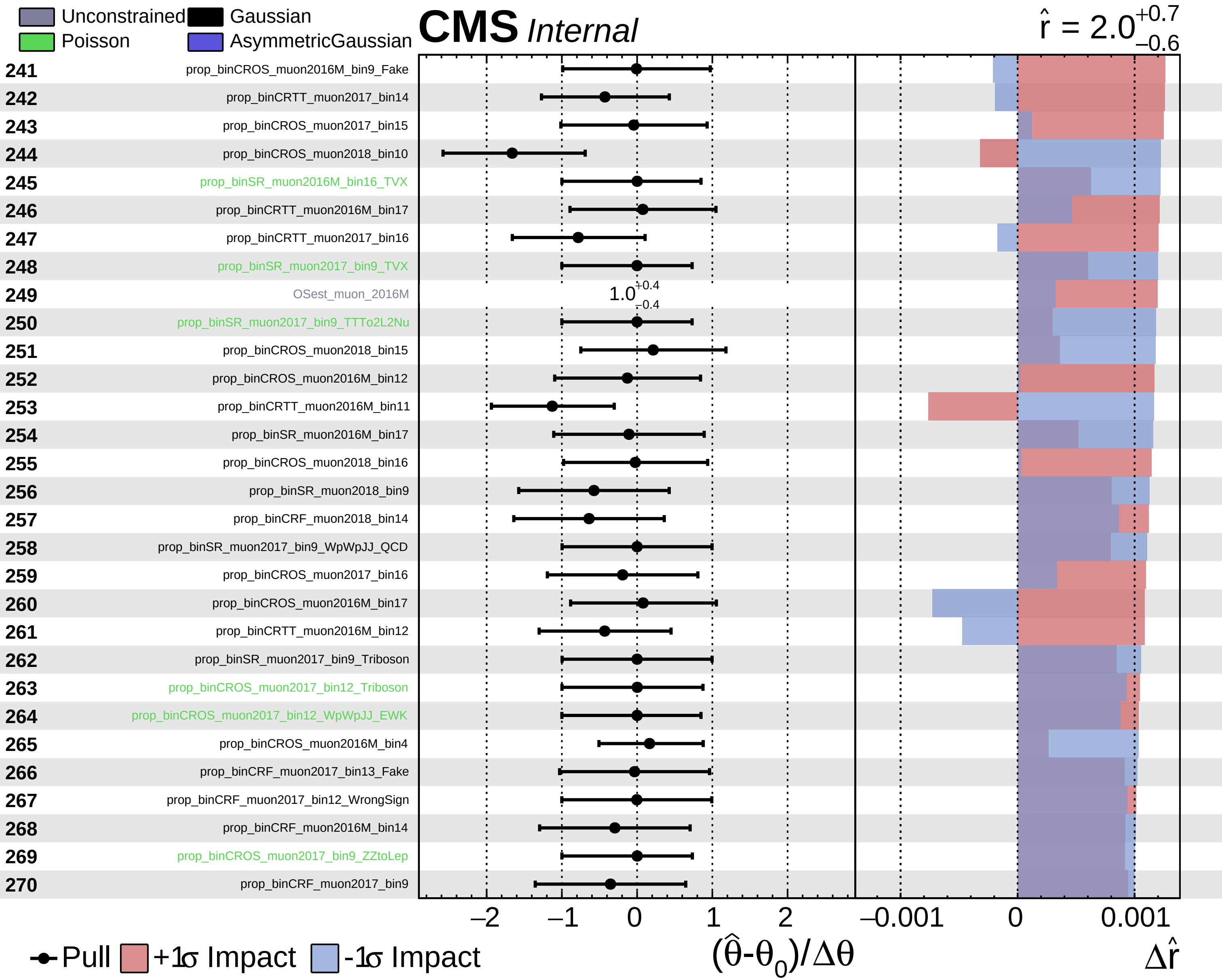

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
151 QCDScale_TVX
152 prop_binCRTT_muon2017_bin17
153 prop_binCROS_muon2016M_bin11
154 prop_binCRTT_muon2018_bin10
155 prop_binCRF_muon2016M_bin9
156 TES
157 prop_binCRTT_muon2018_bin11
158 prop_binCRTT_muon2017_bin5
159 prop_binCROS_muon2017_bin10
160 prop_binCRTT_muon2017_bin6
161 prop_binCROS_muon2017_bin2
162 prop_binCROS_muon2018_bin4
163 prop_binCROS_muon2017_bin17
164 prop_binCRTT_muon2017_bin13
165 prop_binSR_muon2017_bin14_WpWpJJ_EWK
166 prop_binSR_muon2016M_bin16_ZZtoLep
167 QCDScale_Triboson
168 prop_binCROS_muon2016M_bin9_WrongSign
169 prop_binSR_muon2017_bin16_ZZtoLep
170 prop_binCRTT_muon2018_bin3
171 prop_binCRTT_muon2017_bin4
172 prop_binCRTT_muon2017_bin7
173 prop_binCRTT_muon2016M_bin3
174 prop_binCRTT_muon2018_bin0
175 prop_binCRTT_muon2016M_bin4
176 prop_binCROS_muon2017_bin4_WrongSign
177 prop_binSR_muon2017_bin17
178 ISR
179 prop_binCROS_muon2017_bin9_WrongSign
180 prop_binCRTT_muon2017_bin8
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

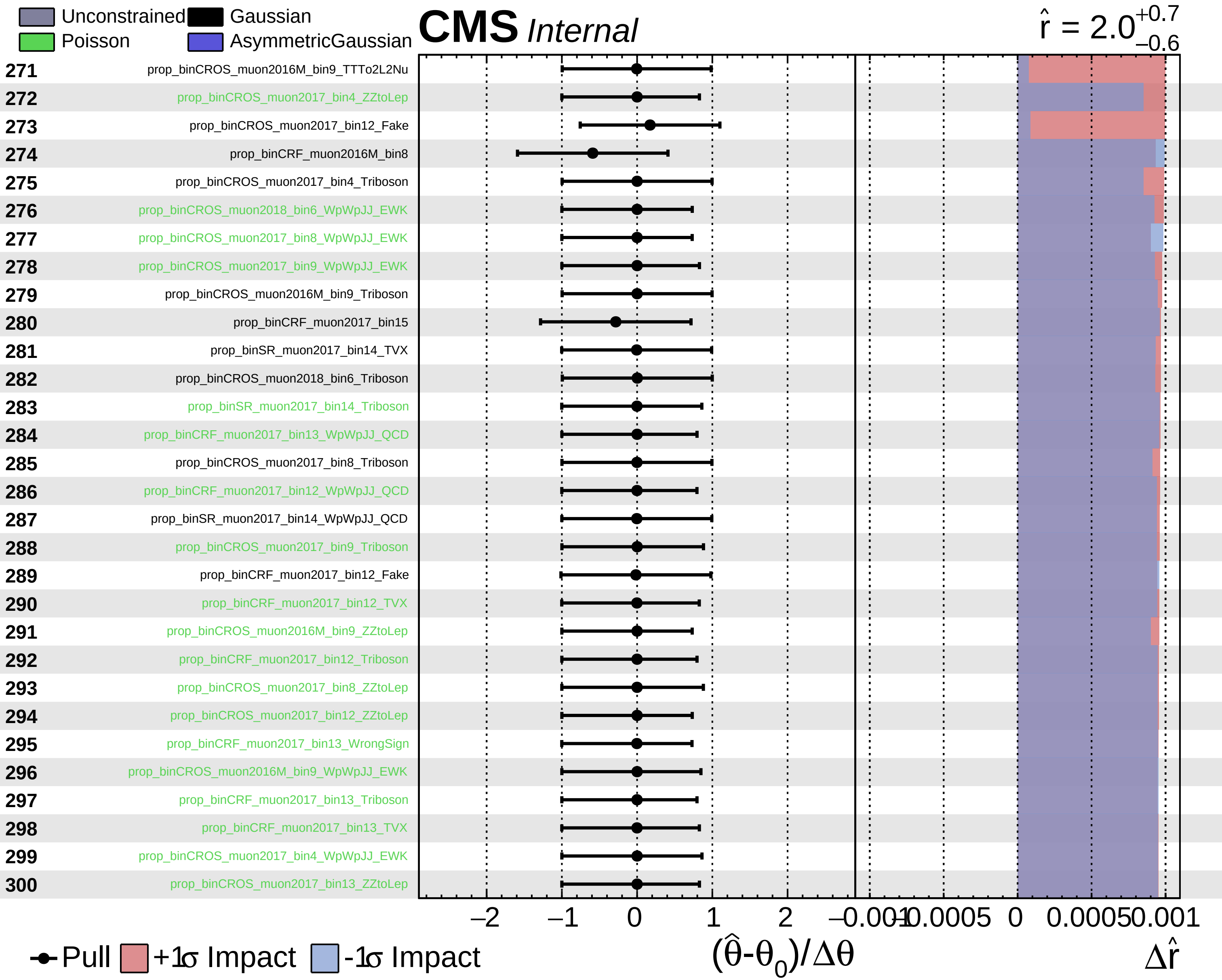


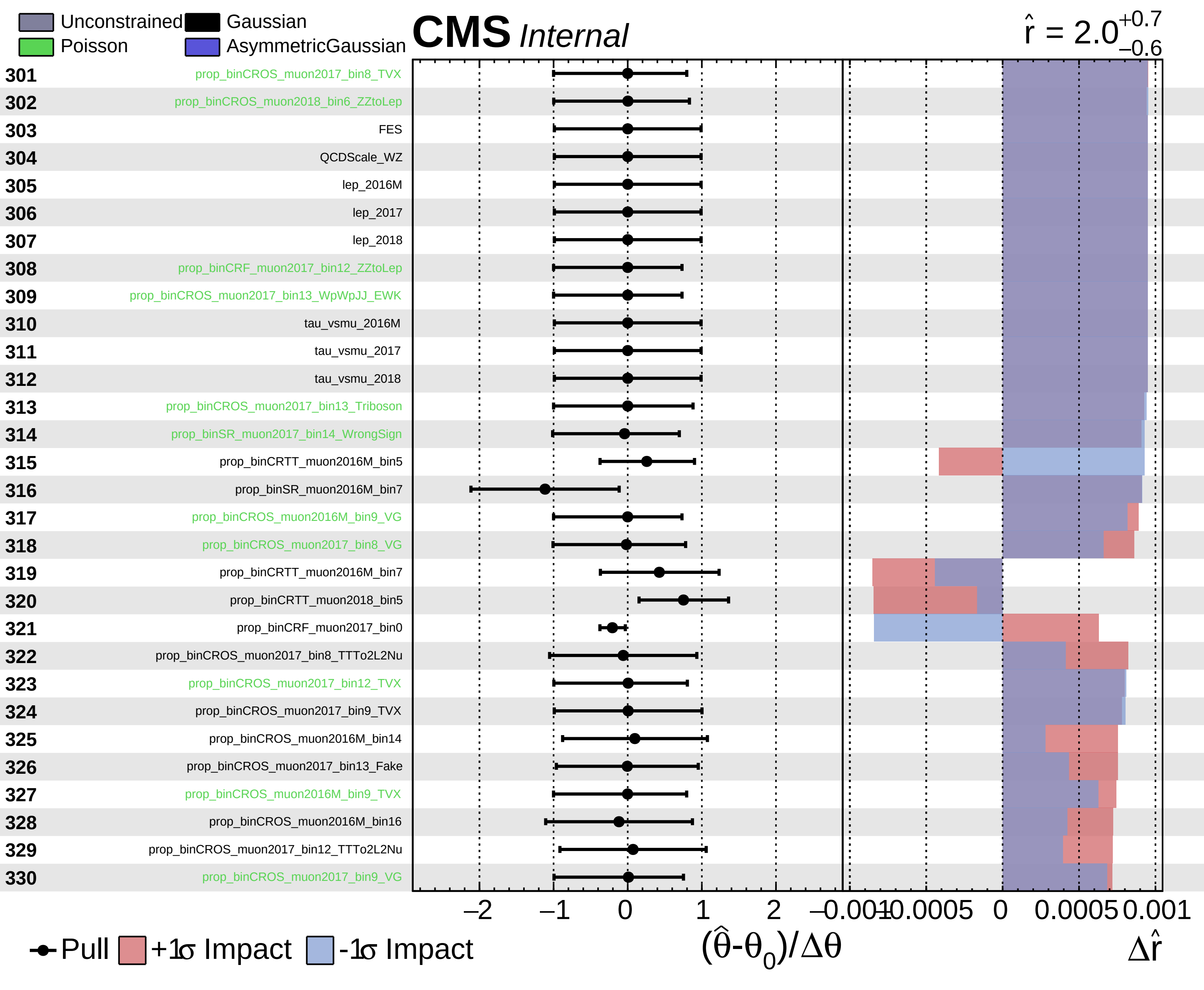
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
181 prop_binCRTT_muon2017_bin11
182 prop_binCROS_muon2017_bin13_WrongSign
183 prop_binCROS_muon2018_bin6_WrongSign
184 prop_binSR_muon2017_bin16_WrongSign
185 prop_binCROS_muon2017_bin12_WrongSign
186 prop_binCROS_muon2018_bin17
187 prop_binSR_muon2017_bin16_WpWpJJ_EWK
188 prop_binSR_muon2017_bin16_WpWpJJ_QCD
189 prop_binCRTT_muon2018_bin12
190 prop_binSR_muon2016M_bin16_WpWpJJ_EWK
191 prop_binCRTT_muon2017_bin0
192 OSest_muon_2017
$0.89^{+0.33}_{-0.32}$
193 prop_binCROS_muon2016M_bin8
194 prop_binCRTT_muon2016M_bin10
195 jer_2016M
196 prop_binCRF_muon2016M_bin0
197 prop_binCROS_muon2017_bin7
198 prop_binCRTT_muon2016M_bin14
199 prop_binCRTT_muon2016M_bin9
200 mischarge_muon_2016M
201 QCDScale_VG
202 prop_binCRF_muon2017_bin12_WpWpJJ_EWK
203 prop_binSR_muon2017_bin16_Triboson
204 prop_binCRF_muon2017_bin12_WZ
205 prop_binCRTT_muon2016M_bin2
206 prop_binCROS_muon2018_bin9
207 prop_binCRTT_muon2018_bin16
208 prop_binCRF_muon2017_bin13_WpWpJJ_EWK
209 prop_binCRTT_muon2017_bin2
210 prop_binCRF_muon2017_bin11
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



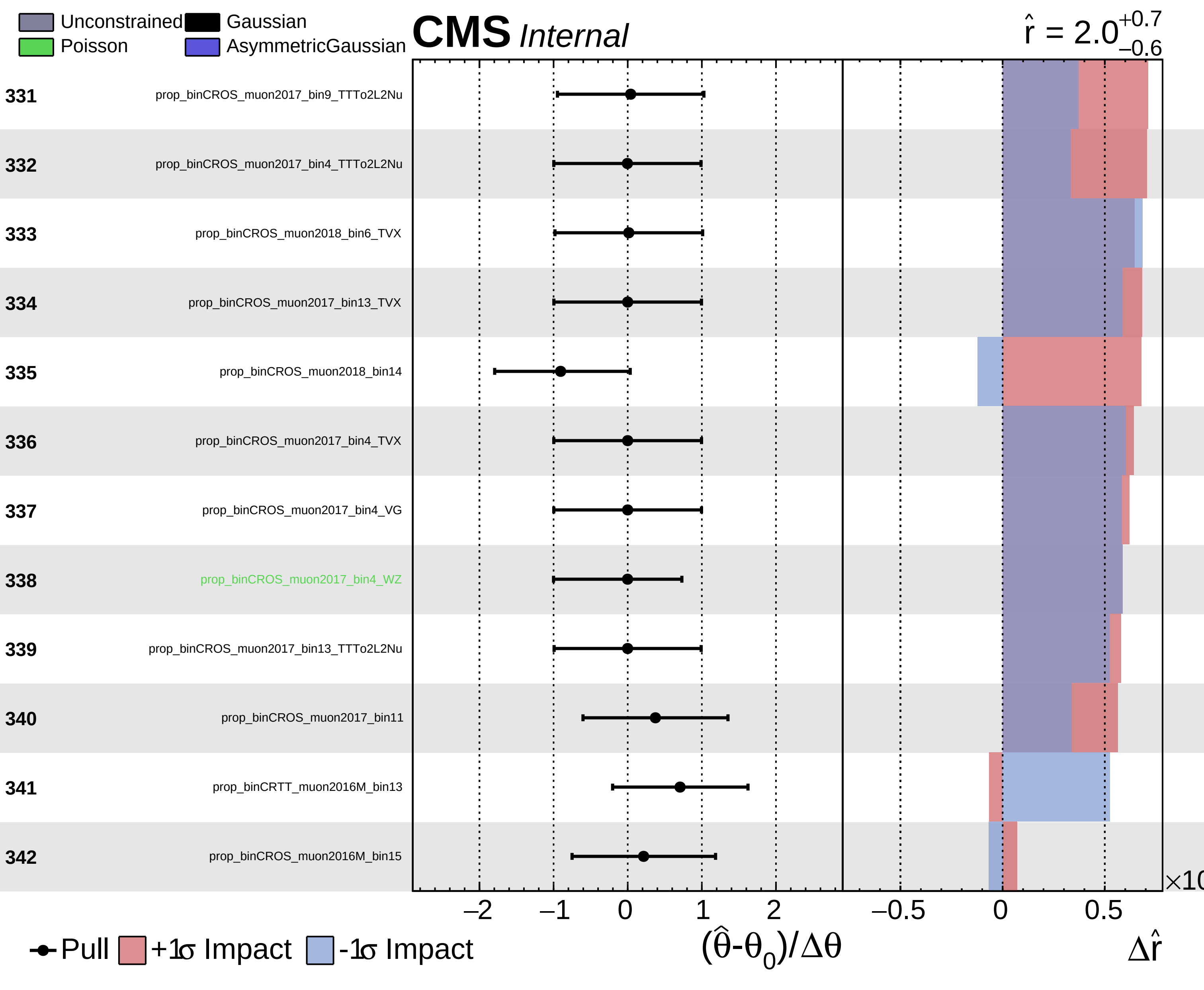
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
211 prop_binCRF_muon2018_bin16
212 prop_binCRF_muon2017_bin14
213 prop_binCRTT_muon2016M_bin8
214 mischarge_muon_2018
215 prop_binCRF_muon2016M_bin12
216 prop_binCRTT_muon2017_bin1
217 prop_binCRTT_muon2018_bin14
218 prop_binCRF_muon2018_bin0
219 prop_binCRTT_muon2016M_bin16
220 prop_binCROS_muon2016M_bin5
221 prop_binCRF_muon2016M_bin15
222 prop_binSR_muon2017_bin9_WpWpJJ_EWK
223 prop_binCRTT_muon2017_bin12
224 prop_binCRTT_muon2016M_bin15
225 prop_binSR_muon2016M_bin15
226 prop_binCROS_muon2016M_bin13
227 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
228 prop_binSR_muon2016M_bin16_WrongSign
229 QCDScale_WrongSign
230 prop_binCROS_muon2017_bin4_Fake
231 prop_binCRTT_muon2017_bin15
232 prop_binCRF_muon2016M_bin16
233 prop_binCRTT_muon2017_bin10
234 prop_binCROS_muon2017_bin8_Fake
235 prop_binSR_muon2017_bin13
236 QCDScale_ZZtoLep
237 prop_binSR_muon2016M_bin16_Triboson
238 prop_binCROS_muon2018_bin6_TTTo2L2Nu
239 prop_binCROS_muon2018_bin13
240 prop_binCROS_muon2017_bin9_Fake
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
241 prop_binCROS_muon2016M_bin9_Fake
242 prop_binCRTT_muon2017_bin14
243 prop_binCROS_muon2017_bin15
244 prop_binCROS_muon2018_bin10
245 prop_binSR_muon2016M_bin16_TVX
246 prop_binCRTT_muon2016M_bin17
247 prop_binCRTT_muon2017_bin16
248 prop_binSR_muon2017_bin9_TVX
249 OSest_muon_2016M
$1.0^{+0.4}_{-0.4}$
250 prop_binSR_muon2017_bin9_TTTo2L2Nu
251 prop_binCROS_muon2018_bin15
252 prop_binCROS_muon2016M_bin12
253 prop_binCRTT_muon2016M_bin11
254 prop_binSR_muon2016M_bin17
255 prop_binCROS_muon2018_bin16
256 prop_binSR_muon2018_bin9
257 prop_binCRF_muon2018_bin14
258 prop_binSR_muon2017_bin9_WpWpJJ_QCD
259 prop_binCROS_muon2017_bin16
260 prop_binCROS_muon2016M_bin17
261 prop_binCRTT_muon2016M_bin12
262 prop_binSR_muon2017_bin9_Triboson
263 prop_binCROS_muon2017_bin12_Triboson
264 prop_binCROS_muon2017_bin12_WpWpJJ_EWK
265 prop_binCROS_muon2016M_bin4
266 prop_binCRF_muon2017_bin13_Fake
267 prop_binCRF_muon2017_bin12_WrongSign
268 prop_binCRF_muon2016M_bin14
269 prop_binCROS_muon2017_bin9_ZZtoLep
270 prop_binCRF_muon2017_bin9
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 2.0 +0.7 −0.6
271 prop_binCROS_muon2016M_bin9_TTTo2L2Nu
272 prop_binCROS_muon2017_bin4_ZZtoLep
273 prop_binCROS_muon2017_bin12_Fake
274 prop_binCRF_muon2016M_bin8
275 prop_binCROS_muon2017_bin4_Triboson
276 prop_binCROS_muon2018_bin6_WpWpJJ_EWK
277 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
278 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
279 prop_binCROS_muon2016M_bin9_Triboson
280 prop_binCRF_muon2017_bin15
281 prop_binSR_muon2017_bin14_TVX
282 prop_binCROS_muon2018_bin6_Triboson
283 prop_binSR_muon2017_bin14_Triboson
284 prop_binCRF_muon2017_bin13_WpWpJJ_QCD
285 prop_binCROS_muon2017_bin8_Triboson
286 prop_binCRF_muon2017_bin12_WpWpJJ_QCD
287 prop_binSR_muon2017_bin14_WpWpJJ_QCD
288 prop_binCROS_muon2017_bin9_Triboson
289 prop_binCRF_muon2017_bin12_Fake
290 prop_binCRF_muon2017_bin12_TVX
291 prop_binCROS_muon2016M_bin9_ZZtoLep
292 prop_binCRF_muon2017_bin12_Triboson
293 prop_binCROS_muon2017_bin8_ZZtoLep
294 prop_binCROS_muon2017_bin12_ZZtoLep
295 prop_binCRF_muon2017_bin13_WrongSign
296 prop_binCROS_muon2016M_bin9_WpWpJJ_EWK
297 prop_binCRF_muon2017_bin13_Triboson
298 prop_binCRF_muon2017_bin13_TVX
299 prop_binCROS_muon2017_bin4_WpWpJJ_EWK
300 prop_binCROS_muon2017_bin13_ZZtoLep
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
301 prop_binCROS_muon2017_bin8_TVX
302 prop_binCROS_muon2018_bin6_ZZtoLep
303 FES
304 QCDScale_WZ
305 lep_2016M
306 lep_2017
307 lep_2018
308 prop_binCRF_muon2017_bin12_ZZtoLep
309 prop_binCROS_muon2017_bin13_WpWpJJ_EWK
310 tau_vsmu_2016M
311 tau_vsmu_2017
312 tau_vsmu_2018
313 prop_binCROS_muon2017_bin13_Triboson
314 prop_binSR_muon2017_bin14_WrongSign
315 prop_binCRTT_muon2016M_bin5
316 prop_binSR_muon2016M_bin7
317 prop_binCROS_muon2016M_bin9_VG
318 prop_binCROS_muon2017_bin8_VG
319 prop_binCRTT_muon2016M_bin7
320 prop_binCRTT_muon2018_bin5
321 prop_binCRF_muon2017_bin0
322 prop_binCROS_muon2017_bin8_TTTo2L2Nu
323 prop_binCROS_muon2017_bin12_TVX
324 prop_binCROS_muon2017_bin9_TVX
325 prop_binCROS_muon2016M_bin14
326 prop_binCROS_muon2017_bin13_Fake
327 prop_binCROS_muon2016M_bin9_TVX
328 prop_binCROS_muon2016M_bin16
329 prop_binCROS_muon2017_bin12_TTTo2L2Nu
330 prop_binCROS_muon2017_bin9_VG
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 2.0^{+0.7}_{-0.6}$
331 prop_binCROS_muon2017_bin9_TTTo2L2Nu
332 prop_binCROS_muon2017_bin4_TTTo2L2Nu
333 prop_binCROS_muon2018_bin6_TVX
334 prop_binCROS_muon2017_bin13_TVX
335 prop_binCROS_muon2018_bin14
336 prop_binCROS_muon2017_bin4_TVX
337 prop_binCROS_muon2017_bin4_VG
338 prop_binCROS_muon2017_bin4_WZ
339 prop_binCROS_muon2017_bin13_TTTo2L2Nu
340 prop_binCROS_muon2017_bin11
341 prop_binCRTT_muon2016M_bin13
342 prop_binCROS_muon2016M_bin15
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$